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An Empirical Study to Detect Cyberbullying with TF-IDF and Machine Learning Algorithms

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Abstract—The headway of technology has enabled various aptitudes like the internet and social media platforms evolving within the internet. Apart from the offerings, there is also the dark side named cyberbullying which is defined as the abusive and personal attacking regarding social interactions on various social platforms. The aftermath of this heinous job is harrowing and tormentous. One momentary solution can evolve the detection of texts regarding cyberbullying. TF-IDF has shown preeminent textual feature map calculation over the years to solve the ambiguity of cross-domain words. Moreover, supervised machine learning algorithms allude to wondrous performance in classifying textual data. Bearing that in mind, a combination of machine learning algorithms namely SVM (Support Vector Machine), Logistic Regression, Decision Tree, Random Forest and MNB (Multinomial Naïve Bayes) has been advocated with TF-IDF features to classify cyberbullying from text data. Two different approaches have been engendered namely the baseline approach that proceeds with imbalanced data and SMOTE (Synthetic Minority Over-sampling Technique) approach for balancing the data where Random Forest with SMOTE approach has achieved an accuracy of 89% with an f1-score of 89% in this empirical study.

Index Terms—Cyberbullying, TF-IDF, SVM, Random Forest, SMOTE.

I. INTRODUCTION

Furtherance of technology has given us enormous facilities evolving the internet and social media platforms. But this rapid growth of the internet has enabled people to misuse it as well. Such a devastating phenomenon is cyberbullying which is termed as aggressive and abusive behavior toward people regarding digital interactions [1]. The consequences of such behavior are also ruinous as they result in mental disorders like depression, trauma, suicide, etc. among victims. Apart from the adult people, teenagers, as well as kids, are also involved in cyberbullying. Online Report of EU Kids of 2014 shows 20% of the children from age 11 to 16 are mainly indulged in cyberbullying whereas among the teenagers the rate is between 20% and 40% [2], [3]. People abuse others on online platforms like Facebook, Twitter, YouTube, etc. by texting abusive comments full of hatred and slander. Introducing an identification system to these texts can stop cyberbullying by taking actions against the menacing act as well as help normal people to get rid of the heinous consequences. TF-IDF (Term Frequency – Inverse Document Frequency) gives momentous values in terms of classifying texts. These values

solve the ambivalence nature of texts in cross-domain by expressing the signification of the words in the document [4]. Moreover, various supervised machine learning algorithms show momentous results pointing to the significance of these models in natural language processing [5]. TF-IDF with the assistance of these machine learning algorithms can uphold portentous outcomes in terms of classifying cyberbullying from text data.

In this study, we have recognized this event of cyberbullying and suggested a method amalgamating TF-IDF and supervised machine learning algorithms like SVM, Logistic Regression, Decision Tree, Random Forest, and MNB to classify cyberbullying from textual data. We have also advocated two approaches in this empirical research namely the baseline approach propagating with the imbalance data and SMOTE (Synthetic Minority Over-sampling Technique) [6] for balancing the data by synthesis.

The other parts of the study are inaugurated as related works and methodology has been embellished in Section II and III respectively ornamenting the proper position of the classifier algorithms. Experimental results and analysis has been delivered in Section IV to manifest the novelty and significance of the study. Finally, the study is terminated with the conclusion and future works in Section V.

II. RELATED WORKS

Ruinous consequences of cyberbullying have both allured and obviated researchers around the world to find notable solutions in terms of its detection. A cyberbullying identification system was purported by authors to prevent cyberbullying in social platforms in their research [7]. TF-IDF features with sentiment analysis had been advocated with SVM classifier by the authors that achieved the accuracy and f1-score of 75.17% and 75% respectively.

A Tweeter-based cyberbullying detection was proposed by authors in their study [8]. TF-IDF had been used as features with machine learning algorithms namely SVM and Naïve Bayes where they achieved 71.25% and 52.70% accuracy respectively. A multimodal approach for cyberbullying detection was introduced by authors [9]. Word tokens with various deep learning models had been used where the Residual BiLSTM-RCNN (Bidirectional Long Short Term Memory - Recurrent

E. Training of Data

The dataset has been split using the percentage splitting method. The ratio of the training and testing set is 80% and 20% respectively.

F. Tools and Simulation Environment

Python 3.8.5, Jupyter Notebook 6.0.3, Pandas 1.0.1, Numpy 1.18.5, Sk-learn 0.24.1, Imb-learn 0.8.0, Matplotlib 3.1.3 and Seaborn 0.10.0 have been used in this study.

G. Classification Models

1) *Support Vector Machine*: Support Vector Machine (SVM) classifier sets support vectors near the boundary of particular classes to categorize instances. The linear kernel of the SVM has been wielded with the assistance of sklearn [15] library.

2) *Logistic Regression*: For the proposed binary classification, Logistic Regression has been advocated with parameters namely tolerance value as 0.0001 and inverse of regularization strength of 1.0.

3) *Decision Tree*: Decision Tree classifies data with the importance of features on entropy or similar metric values. Decision Tree with ‘gini’ criterion parameter as Gini impurity has been espoused in this study.

4) *Random Forest*: Random Forest is a conglomeration of several decision tree organizations. We have advocated Random Forest with 100 as the number of estimators and ‘gini’ criterion for Gini impurity as parameters for the proposed method.

5) *Multinomial Naïve Bayes*: Multinomial Naïve Bayes is a probabilistic learning model that works significantly well with textual data analysis with fractional valued features. Multinomial Naïve Bayes has been utilized with an alpha value of 1 in this regard.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

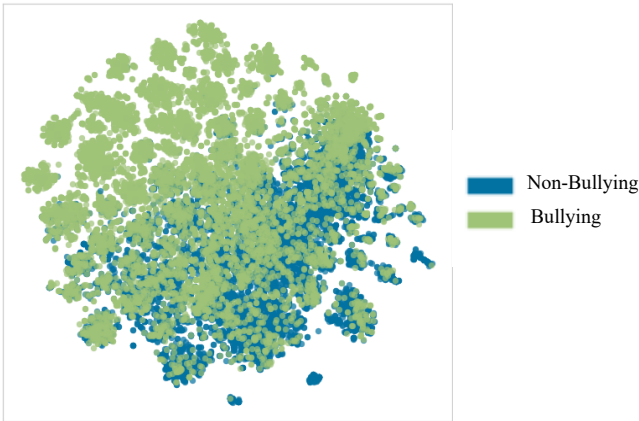


Fig. 4. TF-IDF spread of the dataset.

Fig. 4, espoused by plotting the TF-IDF values of features with the support of Matplotlib library, portrays the spread of TF-IDF values of both the classes of the dataset pigmented

by respective colors and hints at the two clusters in terms of TF-IDF. The shared regions signify that there are words that are cross-domain. Again the lonely scattering TF-IDF values of the bullying class, floating in the upper part of the figure, postulate that there are words that strongly threshold to the bullying class.

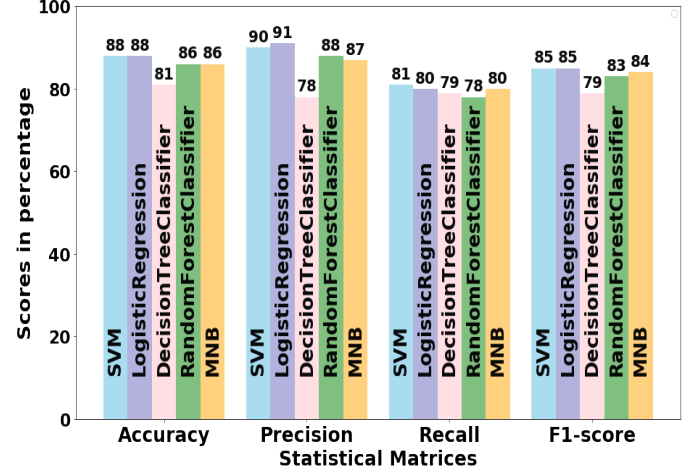


Fig. 5. Evaluation matrices of various models (Baseline approach).

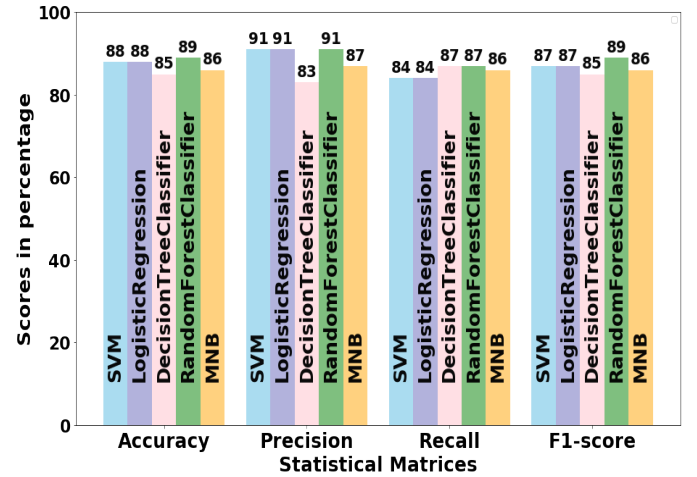


Fig. 6. Evaluation matrices of various models (SMOTE approach).

Fig. 5 and 6 portray the performance of both the baseline approach and the SMOTE approach respectively with statistical matrices on test data. In the baseline approach, SVM and Logistic Regression both have gained the peak accuracy and f1-score of 88% and 85% respectively. In terms of precision, Logistic Regression has gained 91% where SVM has achieved 90%. However, with recall SVM has attained 81% and Logistic Regression 80%. Decision Tree has performed the least in this regard. The SMOTE approach has achieved better results than the baseline approach. The Random Forest has gained an upper hand in all the statistical matrices with accuracy, precision, recall, and f1-score as 89%, 91%, 87%, and 89% respectively. The agglomeration of various Decision

Tree configurations and their ensembled learning has espoused better results with TF-IDF values for Random Forest. Decision Tree has performed the least in this regard.

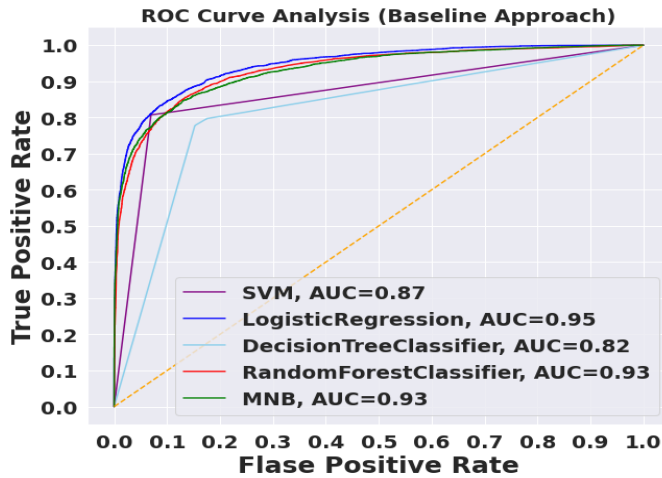


Fig. 7. AUC & ROC Curve (Baseline approach).

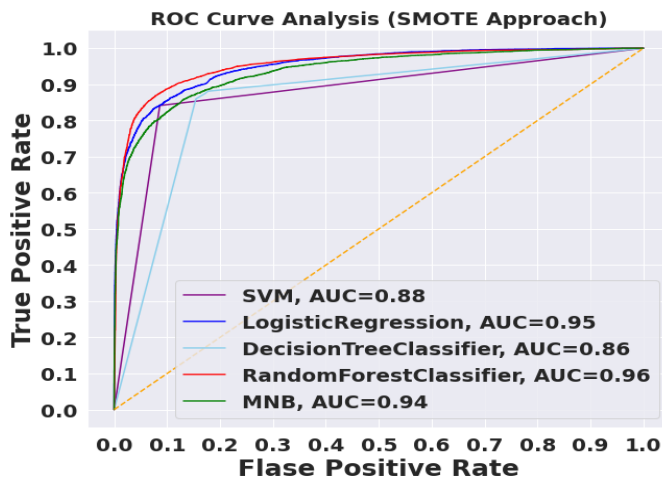


Fig. 8. AUC & ROC Curve (SMOTE approach).

Fig. 7 and 8 illustrate the AUC (Area Under The Curve) and ROC (Receiver Operating Characteristics) curve of the baseline and SMOTE approach respectively. In the baseline approach, the highest AUC of 0.95 has been achieved by Logistic Regression where the least was 0.82 with Decision Tree. The SMOTE approach has performed better in terms of the AUC as well since the Random Forest has gained the highest AUC of 0.96 and Decision Tree the least with 0.86.

V. CONCLUSION AND FUTURE WORKS

Technological advancement has brought revolutionary changes with internet and social media interactions. Such development has also introduced menacing aspects like cyberbullying that involves slander and abusive comments towards people with devastating consequences. Such consequences can be avoided if an identification system is advocated for

cyberbullying detection. TF-IDF has brought new boundaries defining the significance of words in particular domains assisting the classification tasks. Again supervised machine learning algorithms with TF-IDF can be a magnificent combination in terms of text classification. Aligning with that concept, a conglomeration of TF-IDF and supervised machine learning algorithms namely SVM, Logistic Regression, Decision Tree, Random Forest, and MNB has been espoused to categorize cyberbullying from textual data. Two different approaches have been introduced in that regard, the baseline approach with imbalanced data and the SMOTE approach for balancing the data where Random Forest with SMOTE approach has gained an accuracy of 89% with an f1-score of 89%.

Some Unicode remaining undetected have resulted in the removal of some words which is a limitation. Again the size of the dataset and training time is a limitation since the more the data involves it takes more time to train the dataset. In near future, we would like to advocate some other dataset-balancing algorithms. Again we would like to espouse the sequence of words as features with various deep learning algorithms.

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